

Bi-LSTM and Deep Reinforcement Learning Strategy for Long-Term Profit Optimization in the Financial Subgroup of Thailand's SET50 Index Using LLM-Derived Contextual Features

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1 Introduction and Problem Statement

The rapid advancement of machine learning (ML) has introduced data-driven techniques that are capable of adapting to complex financial market dynamics [1]. Deep learning models, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM), have shown superior performance in capturing sequential data like stock prices compared to traditional statistical methods [2], [3]. This capability is further enhanced by the use of **Bi-Directional Long Short-Term Memory (Bi-LSTM)**, which analyzes input data in both forward and backward directions, thereby improving prediction accuracy and offering a more comprehensive understanding of market trends [2], [4], [5], [6].

The original proposal addresses a research gap where current models often focus on short-term trading or rely on limited data types, which weakens robustness in real-world, long-term investment strategies within Thailand's SET50 Index [4], [7], [6], [8]. This project leverages the complexity of Bi-LSTM combined with **multi-source data integration** to develop a robust, adaptive trading strategy focused on long-term profit maximization [9], [10], [11].

The project has been enhanced to address complexity and accuracy by:

1. **Narrowing the Scope:** Focusing the analysis on **5-10 major financial stocks** within the SET50 Index (based on conversation history) [4].
2. **Upgrading Feature Engineering (NLP/LLM):** Moving beyond basic keyword-based sentiment scoring [12] to utilize a **full Large Language Model (LLM)** approach for generating sophisticated, contextual risk and event features from financial news (based on conversation history) [4].
3. **Shifting to Decision-Making (DRL):** Implementing a **Deep Reinforcement Learning (DRL) agent** to optimize the trading strategy directly, maximizing risk-adjusted financial returns rather than focusing solely on return forecasts (based on conversation history) [4].

2 Objectives

The refined objective is to design, implement, and validate a highly adaptive trading strategy that maximizes long-term, risk-adjusted profitability in the chosen financial subgroup of the SET50 Index [5].

The specific objectives are:

- To design and build an advanced feature extraction mechanism using an **LLM-based model** to generate contextual risk and event feature vectors from multi-source data (Thai financial news, Google News API) for the selected 5-10 major financial stocks [5].
- To integrate the LLM-derived features and Bi-LSTM time-series features into the state space of a **Deep Reinforcement Learning (DRL) agent** [5].
- To train the DRL agent to execute an optimal trading sequence aimed at maximizing key financial performance metrics, such as the **Sharpe Ratio and Sortino Ratio**, over a ten-year backtesting period [5], [13].

3 Methodology and Data Sources

The project will use multi-source data collected over a 10-year span [12], [13].

Table 1: Project Data Sources

Data Source Category	Specific Inputs	Citation
Price History	OHLCV data for 5-10 major financial stocks (subset of SET50 Index)	[14]
Technical Indicators	SMA, EMA, RSI, MACD, Bollinger Bands, and Stochastic Oscillator	[14]
Contextual/Behavioral Data	LLM-Derived Risk/Event Vectors from Thai financial news headlines and Google News API	[14], (Enhanced from keyword scoring [12])
Macroeconomic Indicators	Exchanged rates and global indices	[14]

3.1 Key Procedures

1. **Feature Engineering (LLM/Bi-LSTM):** The LLM component will process textual data to extract contextual features, generating highly relevant "sentiment signals" [10], [14]. The Bi-LSTM model will process sequential data (price and technical indicators) to capture long-term temporal dependencies [3].
2. **Model Architecture (DRL):** The DRL agent will utilize the combined output of the LLM and Bi-LSTM as its state vector input to make discrete or fractional

trading decisions (actions). This shifts the primary focus from continuous return forecasts [6] to direct decision-making [9].

3. **Strategy Simulation:** The DRL agent’s policy will be optimized within a simulated environment focused on **long-term profit maximization** [10].
4. **Evaluation and Optimization:** Performance will be rigorously tested via back-testing (e.g., training 2014–2021, testing 2022–2024) [13]. Hyperparameter optimization will use Grid search or Bayesian optimization with walk-forward validation [13], [15].
5. **Benchmarking:** The DRL strategy will be benchmarked against baseline models, including traditional ML models (Random Forest, XGBoost), technical-rule-based strategies, and the passive buy-and-hold strategy [15].

4 Computational Resource Utilization (KVIS Mini-HPC Cluster)

The integration of advanced deep learning (Bi-LSTM, LLM feature extraction) and Deep Reinforcement Learning (DRL) requires substantial computational power, which will be provided by the **KVIS Mini-HPC Cluster (BistKA)** [8]. The cluster is designed to support computationally demanding research projects, utilizing CPU, GPU, and high-memory resources [8].

Table 2: BistKA Resource Requirements and Usage Policy

Component	BistKA Resource Requirement	Specific Use/Policy
LLM and DRL Neural Network Training	GPU Accelerators	Training Transformer-based models (LLM feature extraction) and DRL neural components requires the RTX PRO 4500 GPUs [11], [16]. Jobs must be submitted to GPU partitions (e.g., <code>gpu4500-normal</code> , 2 days walltime) [17], [18]. The group limit is 2 GPUs [17].
DRL Environment Simulation	High-Core CPU Compute Nodes	DRL agents rely on massive parallel simulations. Jobs should utilize high-core compute nodes (e.g., <code>bistka-compute-01</code> with 64 cores/512 GiB RAM) [19] by requesting resources (max 32 CPU cores/group) via SLURM [17].

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Table 2: BistKA Resource Requirements and Usage Policy (continued)

Component	BistKA Resource Requirement	Specific Use/Policy
Software Management	Miniforge3/SLURM	Python/ML frameworks must be installed in isolated environments using Miniforge3 (which provides conda/mamba) after loading the appropriate module [20], [21]. All execution must occur inside SLURM jobs; heavy computation is prohibited on the master node [22], [23].
Initial Resources	Credit and Storage	The project is allocated an initial credit of 10,000 SHr [22], [24]. Multi-source data storage is limited to **300 GiB** in the project directory [25]. Users can check resource usage with the \$ mycredit and \$ myquota commands [26], [27].

5 Expected Outcomes and Evaluation

The integration of LLM features and DRL is expected to **effectively improve long-term profit optimization** [28]. The project will demonstrate that combining technical and behavioral (LLM-derived) data enhances the model’s ability to extract complex signals from various data domains, resulting in a more predictive and adaptive strategy for volatile emerging markets like Thailand’s SET50 Index [29], [30].

The success of the strategy will be validated by superior performance in terms of drawdown control and **risk-adjusted returns** (Sharpe Ratio and Sortino Ratio) compared to baseline models [31]. The final output will serve as a foundation for developing semi-automated trading tools and decision support systems in local financial markets [28], [31].

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